

Celestial Mechanics

Problem Set

fizicyst

This problem set accompanies the *Celestial Mechanics* handout. Section A has five objective (multiple-choice) questions; Section B has three subjective questions, pitched a little harder than Section A. Use $G = 6.674 \times 10^{-11} \text{ N m}^2\text{kg}^{-2}$, $M_{\odot} = 1.989 \times 10^{30} \text{ kg}$, and $1 \text{ AU} = 1.496 \times 10^{11} \text{ m}$ where needed, unless a question tells you otherwise.

Section A — Objective

O1. A planet moves on an elliptical orbit of eccentricity $e = 0.6$ under the Sun's gravity. If its angular momentum (per unit mass, about the Sun) is $\ell = L/m$, its areal velocity dA/dt is:

- (a) ℓ
- (b) $\ell/2$
- (c) 2ℓ
- (d) $\ell/4$

O2. As $r \rightarrow 0$, the effective potential $V_{\text{eff}}(r) = \frac{L^2}{2mr^2} - \frac{GMm}{r}$ for a planet with nonzero angular momentum L behaves as follows:

- (a) $V_{\text{eff}} \rightarrow -\infty$, so the planet can collide with the central mass
- (b) $V_{\text{eff}} \rightarrow +\infty$, so the planet is always turned back before reaching $r = 0$
- (c) $V_{\text{eff}} \rightarrow 0$, so angular momentum has no effect at small r
- (d) V_{eff} oscillates in sign depending on e

O3. An asteroid orbits the Sun with semi-major axis $a = 4 \text{ AU}$. Taking Earth's period as 1 yr at 1 AU, Kepler's Third Law gives the asteroid's orbital period as:

- (a) 2 yr
- (b) 4 yr
- (c) 8 yr
- (d) 16 yr

O4. Which one of the following statements about the Lagrange points is **false**?

- (a) L_1, L_2, L_3 are unstable equilibria, regardless of the mass ratio M_1/M_2 .
- (b) L_4 and L_5 are stable equilibria for *any* mass ratio M_1/M_2 .
- (c) L_4 and L_5 sit at the apex of equilateral triangles built on the M_1 – M_2 separation.
- (d) Stability of L_4 and L_5 requires $M_1/M_2 \gtrsim 24.96$.

O5. A satellite happens to have exactly the same mean density as its parent planet ($\rho_s = \rho_M$). Its Roche limit d_R , expressed in units of the planet's radius R , is closest to:

- (a) $0.79 R$
- (b) $1.26 R$
- (c) $2.00 R$
- (d) $1.00 R$

Section B — Subjective

S1. A comet orbits the Sun with perihelion distance $r_p = 0.5$ AU and aphelion distance $r_a = 30$ AU.

- (i) Find the semi-major axis a and eccentricity e of the comet's orbit.
- (ii) Using the vis-viva equation, find the comet's speed at perihelion and at aphelion.
- (iii) Find the orbital period T of the comet, in years.

S2. A planet orbits the Sun (mass M) with specific angular momentum $\ell = L/m$ and specific mechanical energy $\varepsilon = E/m$, both conserved along the orbit. At perihelion (r_p) and aphelion (r_a), the planet's velocity is purely tangential, so $v = \ell/r$ at each of these two points.

- (i) Write ε in terms of ℓ , r , G , and M only, and apply this once at r_p and once at r_a .
- (ii) By eliminating ε between your two equations from (i), show that $\ell^2 = \frac{2GM r_p r_a}{r_p + r_a}$. Then, using $r_p = a(1 - e)$, $r_a = a(1 + e)$, and $\varepsilon = -GM/2a$, show that

$$e = \sqrt{1 + \frac{2\varepsilon\ell^2}{G^2M^2}}.$$

- (iii) Check your result against the special case of a circular orbit: using $v^2 = GM/r$ directly, find ℓ and ε for a circular orbit, and confirm your formula from (ii) correctly gives $e = 0$.

S3. Saturn has mass $M = 5.68 \times 10^{26}$ kg, radius $R = 6.03 \times 10^7$ m, and mean density $\rho_M \approx 687$ kg/m³. Consider one of its icy moons, of mean density $\rho_s = 920$ kg/m³.

- (i) Compute the Roche limit d_R for this moon.
- (ii) Suppose the moon originally orbited at $d = 3R$ (well outside d_R) and gradually spiraled inward. Using Kepler's Third Law, estimate the orbital period at the Roche limit itself, taking the total mass to be approximately M (the moon's own mass is negligible).
- (iii) Using the barycentre concept, briefly justify why treating the moon's mass as negligible is reasonable here, and state the condition under which this approximation would break down.